# Borhaneddine Hamadou

Research Engineer (R&D) | Safety Validation & Testing of Autonomous Systems

+33 7 84 34 66 86 | hamadou.borhaneddine@gmail.com | linkedin.com/in/borhaneddine-hamadou | github.com/BorhaneddineHamadou

borhaneddinehamadou.github.io | Île-de-France, France

## Summary

PhD researcher in **safety validation and testing of autonomous driving systems** at CEA List (Université Paris-Saclay), defending July 2027. Three years building testing methods — scenario generation, test adequacy metrics, regression test selection — grounded in **ISO 21448 (SOTIF)** and **ISO 34503** and validated on production driving stacks (Autoware, openpilot) in **CARLA** and **MetaDrive**. Published at **ASE** and **ISSRE**, and shipped as open-source tooling. Previously machine learning for autonomous-vehicle edge computing at **LAAS-CNRS**.

## Education

**Université Paris-Saclay — CEA List** — Palaiseau, France
*PhD in Computer Science — Safety Testing of Autonomous Vehicle Systems* — *Jul 2024 — Jul 2027 (expected)*

- Thesis on scenario-based testing and safety validation of autonomous driving systems, combining Bayesian networks, combinatorial coverage, and statistical modelling.
- Methods evaluated on open-source production stacks (Autoware, openpilot) in CARLA and MetaDrive simulation, against ISO 21448 (SOTIF) and ISO 34503 requirements.

**École Nationale Supérieure d'Informatique (ESI)** — Algiers, Algeria
*Master of Engineering (Ingénieur d'État) in Computer Systems Engineering* — *Sep 2018 — Sep 2023*

- Leading computer science school in Algeria. Specialisation in machine learning, combinatorial optimisation, data analysis, high-performance computing, distributed systems, and network and systems security.
- Engineering thesis (PFE) carried out at LAAS-CNRS, Toulouse — see Professional Experience.

## Professional Experience

**CEA List** — Palaiseau, France
*PhD Researcher (R&D) — Autonomous Systems Testing* — *Jul 2024 — Present*

- Designed **PWCC**, a risk-informed test adequacy metric weighting combinatorial coverage by operational probability learned from fleet data: the **strongest predictor of fault detection** across 4 autonomous driving datasets and 7 baselines (Spearman $r_s = 0.981$), removing the false "adequate" verdicts behind which standard coverage hides up to **95.8%** of failure-triggering conditions. *Published at ASE 2026.*
- Built **BayScen**, an ISO 34503-grounded scenario generator pairing capability abstraction with Bayesian networks learned from real-world data: critical scenarios $\times$**7 more likely to be physically plausible** than adversarial search, at the highest ODD coverage among the methods compared. *Published at ISSRE 2026.*
- Released PWCC as a peer-reviewed replication package — CI-ready command-line tool, Docker image, 31 automated tests, fully pinned dependencies — archived on Zenodo under CC BY 4.0 and reproducible end to end in under 30 minutes.
- Implemented the full experimental pipeline in Python (pgmpy, scikit-learn, NumPy, pandas) over large-scale simulation campaigns.

**LAAS-CNRS** — Toulouse, France
*AI Research Intern — Autonomous Vehicles & Edge Computing* — *Oct 2022 — Sep 2023*

- Engineered time-series forecasting models predicting Quality-of-Service degradation for autonomous vehicles in dynamic edge-computing environments, enabling intervention *before* latency incidents occur.
- Delivered the models as a real-time prediction REST API, integrated into a larger autonomous-vehicle ecosystem for direct deployment.

**Université Paris-Saclay** — France
*Teaching Assistant — Computer Science* — *Jan 2025 — Jul 2026*

- Laboratory and tutorial sessions in Object-Oriented Programming (Java) and Database Management Systems (Oracle: relational design, SQL, PL/SQL, administration), including supervision of student projects.

**Software Developer (Freelance)** — Somu Raqami, Algeria · Full-stack mobile & web — Apr 2019 — Sep 2022
**Software Developer Intern** — Maystro Delivery, Algiers, Algeria · Stock, sales & POS web app — Jul — Sep 2021

## Selected Projects

**PWCC — Test Adequacy Tool for Autonomous Systems** *Python, pgmpy, Docker, pytest · CEA List, 2025–2026*

- Open-source command-line tool that learns an operating-domain model from a user's own scenario data, reports which operating conditions still need testing and how much exposure each requires, and exits non-zero when testing is inadequate — usable directly as a CI stopping gate. doi.org/10.5281/zenodo.21739211

**QoS Prediction API for Autonomous Vehicles** *Python, TensorFlow, scikit-learn, REST · LAAS-CNRS, 2022–2023*

- End-to-end machine-learning pipeline and production-style prediction service for Quality-of-Service degradation in autonomous-vehicle edge environments. github.com/BorhaneddineHamadou/QoS_Pred_API_V2

**High-Performance Solver & Hybrid Metaheuristics** *C, CUDA, OpenMP, MPI · ESI Algiers*

- Sequential and parallel CUDA/OpenMP Gauss-Jordan linear solvers benchmarked under high-computation loads; exact, heuristic, metaheuristic, and hybrid solvers for the unbounded knapsack problem.

## Publications

- **B. Hamadou**, C. Mraidha, N. Matragkas. "PWCC: Probability-Weighted Combinatorial Coverage for Risk-Informed Scenario Testing of Autonomous Systems." *ASE 2026 — 41st IEEE/ACM International Conference on Automated Software Engineering (Research Track)*, Munich, Germany.
- **B. Hamadou**, C. Mraidha, N. Matragkas. "BayScen: Standards-Grounded Scenario Generation for Autonomous Vehicle Testing via Bayesian Networks and Capability Abstraction." *ISSRE 2026 — IEEE International Symposium on Software Reliability Engineering (Research Track)*.
- **B. Hamadou**. "Bayesian Network-Based Scenario Generation for Autonomous Systems Testing." *ISSRE 2025 — IEEE International Symposium on Software Reliability Engineering (Doctoral Symposium)*, São Paulo, Brazil.

## Certifications & Training

- **ISTQB Certified Tester Foundation Level** — training course, G2C/GTEC, Tunis, Tunisia (Oct 2024). Test process and test levels, static testing, black-box and white-box test design techniques, test management.
- **Parcours DRIVE® — R&D en Entreprise**, Université Paris-Saclay (2025–2026). Doctoral programme on industrial R&D careers, concluded by a three-day corporate-strategy hackathon at CentraleSupélec.
- **ESSAI 2025 — European Summer School on Artificial Intelligence**, Bratislava, Slovakia (Jul 2025).
- **Formation Ingénieur Entreprendre** — entrepreneurship diploma, Maison de l'Entrepreneuriat, ESI Algiers (2022–2023).

## Technical Skills

**Testing & Validation** Scenario-based testing · Combinatorial testing · Test adequacy & coverage metrics · Regression test selection · Verification & validation (V&V) · ISO 21448 (SOTIF) · ISO 34503 (ODD) · ISTQB foundations

**Simulation & Robotics** CARLA · MetaDrive · Gazebo · ROS 2 · Autoware · openpilot

**ML & Statistics** Bayesian networks (pgmpy) · Probabilistic modelling · Monte Carlo methods · Time-series forecasting · Supervised & unsupervised learning · scikit-learn · TensorFlow · NumPy · pandas

**Programming** Python · C · C++ · Java · SQL / PL/SQL · Bash · JavaScript

**HPC & Parallel** CUDA · OpenMP · MPI · Pthreads

**Tooling** Linux · Git · Docker · pytest · Continuous integration · LaTeX

## Awards & Academic Service

- **Best Product Award ("Product of the Year")** — INJAZ Algeria national competition for young entrepreneurs, 2022.
- **5th place** — Algerian Collegiate Programming Contest (ICPC Foundation), 2022.
- **Organising committee**, JITA 2025 — Junior Conference on Informatics: Theory and Applications, Université Paris-Saclay: communication, visual design, website, and review process.

## Languages

**Arabic** (native) | **French** (full professional proficiency — language of my teaching and of regular research presentations) | **English** (full professional proficiency — research published and presented internationally) | **German** (A2)

References available upon request.